

Store IT:

Science Lab Inventory System

SPP6: Medium-Fi Prototype

Cristina G., Wasani C., Amber G.

CMPS3141 Human Computer Interfaces
University of Belize

Value Proposition, Problem & Solution

Know where every lab item is before you step into the storeroom.

Problem

Students waste time hunting through several storerooms for misplaced or used-up supplies, and supervisors learn about shortages too late.

Solution

Store IT keeps every item's record in one app, so anyone can find an item (by tab, search or voice), update it, and see what is low or empty before supplies run out. Our full design adds QR codes on containers; this prototype covers lists, stock status, voice search and the stock log.

Task 1 · Simple

Find an item and where it is

A lab user needs to know whether a specific chemical or item is available and exactly where it is stored, without checking several rooms.

What changed and how it works

Goal unchanged from SPP4 and SPP5. In the prototype, the user opens the right tab (Equipment, Chemicals or Glassware), or finds the item with search or voice search.

Task 2 · Moderate

Keep the inventory accurate after using an item

After taking, using or returning an item, the user records what changed so the next person sees the true quantity.

What changed and how it works

Combines the SPP4 task 'keep inventory accurate' with the SPP5 task 'record quantity removed'. In the prototype, the user edits the item with the pencil icon, and the stock log shows what was received or used, by whom and why.

Task 3 · Complex

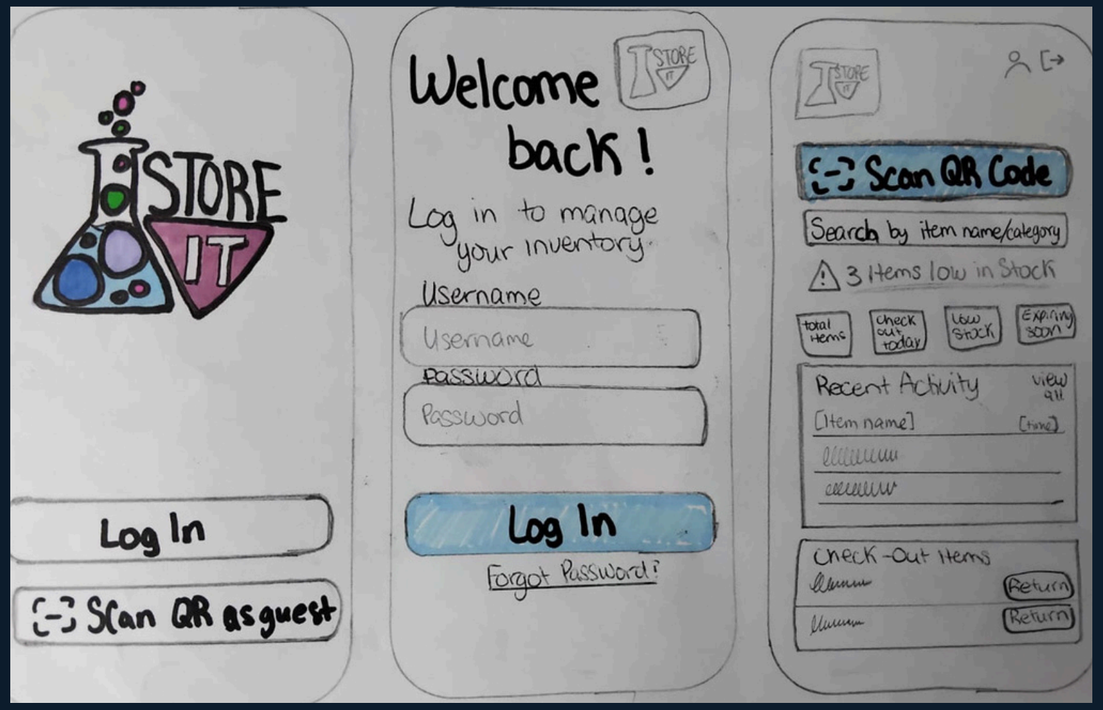
Notice low stock and restock it

A lab user or technician spots that a supply is running low and gets it back on the shelf with a full record, before anyone runs out.

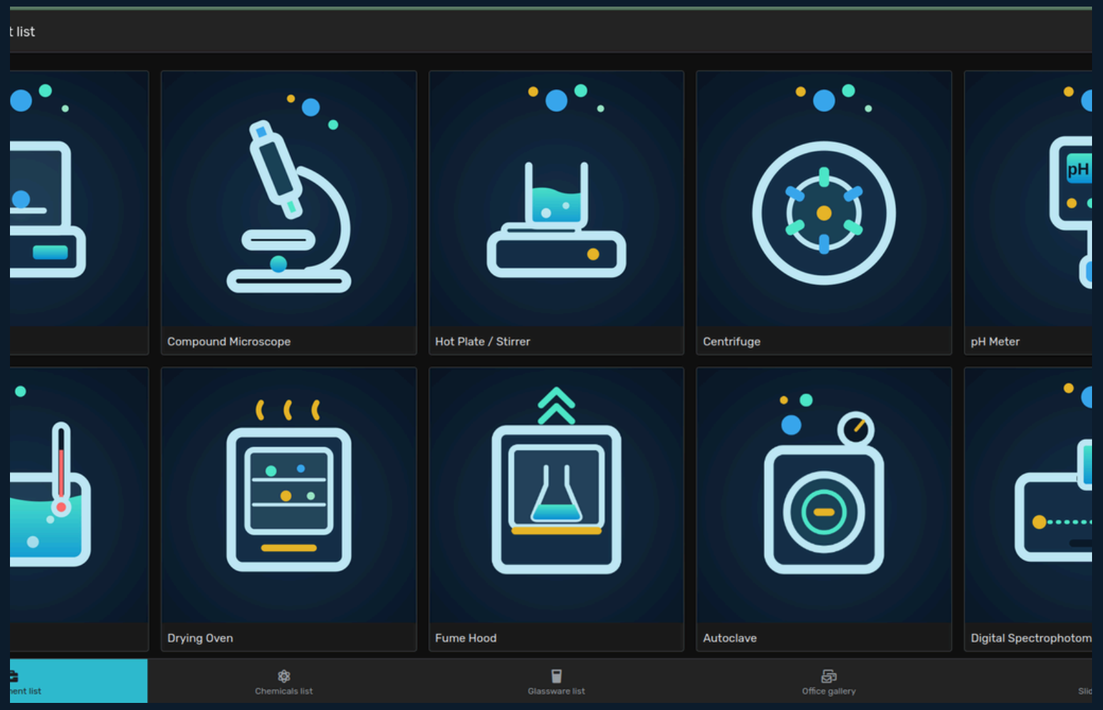
What changed and how it works

Combines SPP4 'restock' with SPP5 'add a new chemical'. In the prototype, each chemical shows whether it is In Stock, Low Stock, Out of Stock or Expired, and new stock is added with the + button.

Change 1: Category tabs instead of one dashboard



Before: one dashboard with a Scan button and stats

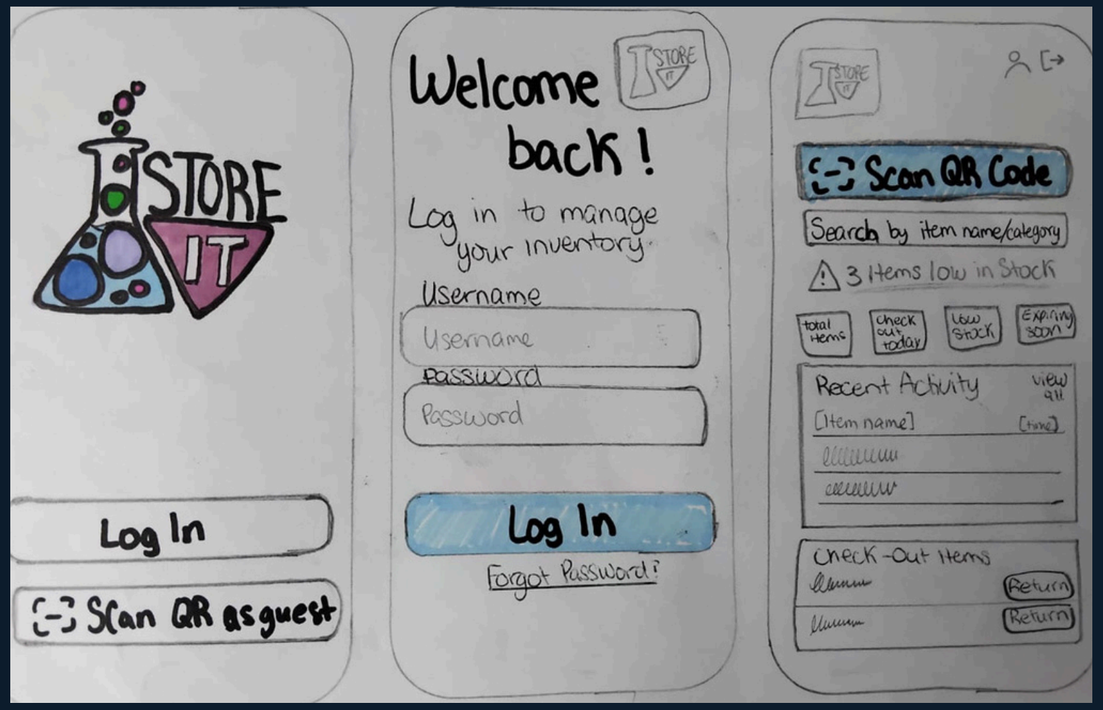


After: AppSheet prototype

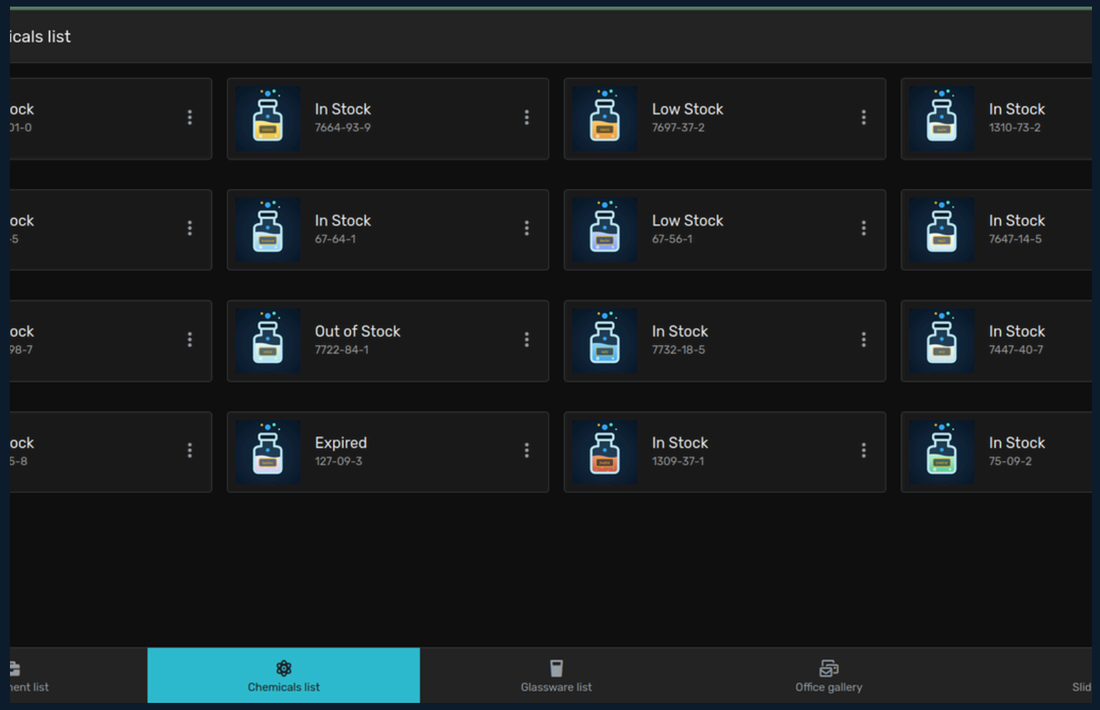
Before, we planned to have everything sit on one dashboard. Now, separate tabs hold Equipment, Chemicals, Glassware, Office supplies and Slides.

Why: pilot testers said reusable glassware stays in fixed places, while chemicals are checked out when used.

Change 2: Stock status on every chemical



Before: one '3 items low in stock' banner

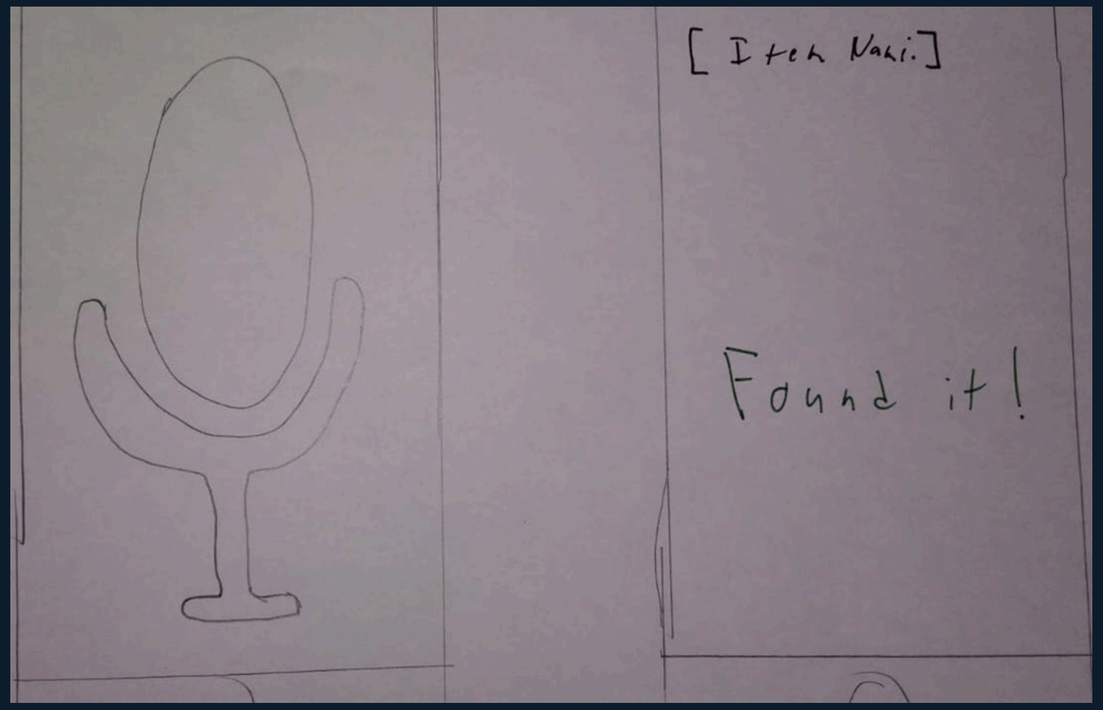


After: AppSheet prototype

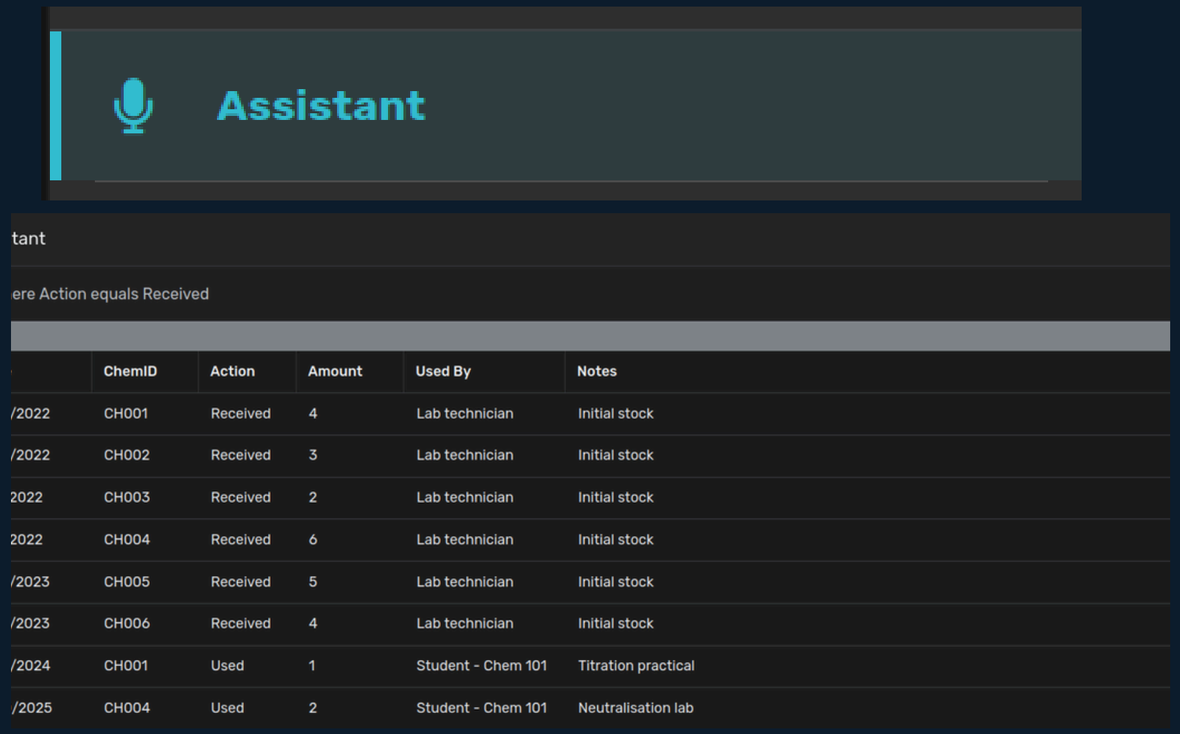
Before, a single banner said how many items were low. Now, each chemical shows its own status: In Stock, Low Stock, Out of Stock or Expired.

Why: restocking (Task 3) needs staff to see which items need attention at a glance.

Change 3: Voice search to find items



Before: early concept sketch of voice search



	ChemID	Action	Amount	Used By	Notes
/2022	CH001	Received	4	Lab technician	Initial stock
/2022	CH002	Received	3	Lab technician	Initial stock
2022	CH003	Received	2	Lab technician	Initial stock
2022	CH004	Received	6	Lab technician	Initial stock
/2023	CH005	Received	5	Lab technician	Initial stock
/2023	CH006	Received	4	Lab technician	Initial stock
/2024	CH001	Used	1	Student - Chem 101	Titration practical
/2025	CH004	Used	2	Student - Chem 101	Neutralisation lab

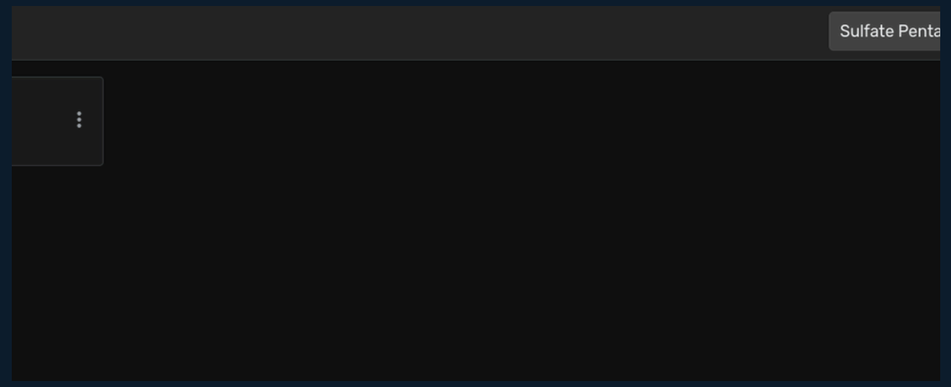
After: Users can search via voice along with typing.

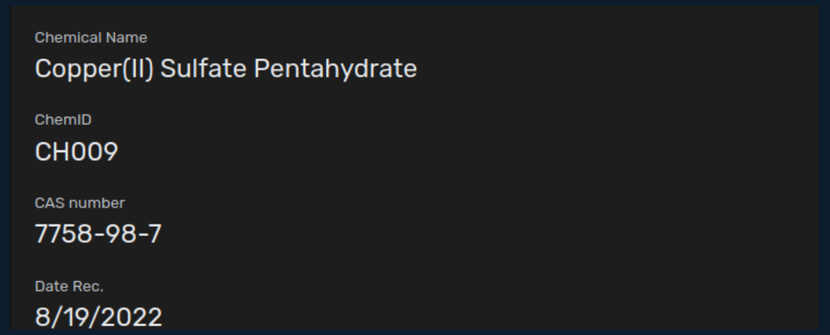
Before, finding an item relied on a Scan QR button. Now, users can also find items with the tabs, by typing, or by speaking the item name.

Why: staff often have their hands full, and speaking is faster than typing.

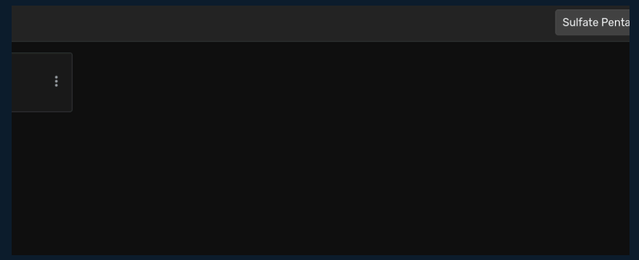
Task Flows in the Medium-Fi Prototype

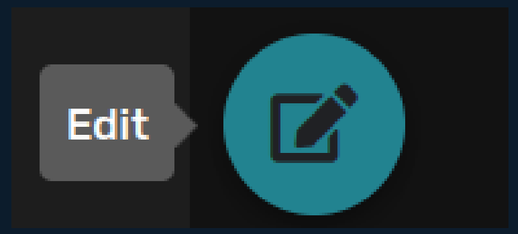
Simple

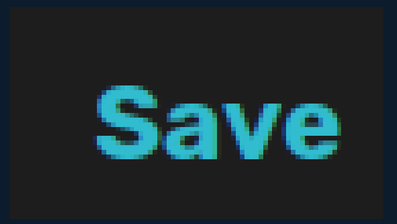




Moderate

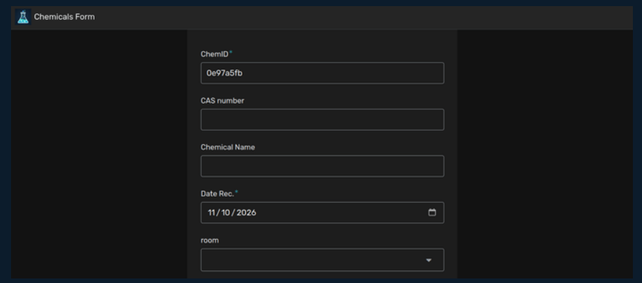


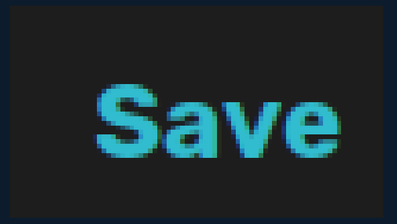




Complex







Prototype Tool: AppSheet

What helped

Quick to build: screens come straight from data tables, with tabs, lists, search, voice search and add/edit built in. It runs in a phone or computer browser.

What was hard

Layouts follow AppSheet's built-in views, so we could not recreate our sketched dashboard exactly. Some features in our design, such as QR scanning, were not built in the time available.

Prototype Tool: AppSheet

What helped

Quick to build: screens come straight from data tables, with tabs, lists, search, voice search and add/edit built in. It runs in a phone or computer browser.

What was hard

Layouts follow AppSheet's built-in views, so we could not recreate our sketched dashboard exactly. Some features in our design, such as QR scanning, were not built in the time available.

Prototype Tool: AppSheet

What helped

Quick to build: screens come straight from data tables, with tabs, lists, search, voice search and add/edit built in. It runs in a phone or computer browser.

What was hard

Layouts follow AppSheet's built-in views, so we could not recreate our sketched dashboard exactly. Some features in our design, such as QR scanning, were not built in the time available.

Prototype Tool: AppSheet

What helped

Quick to build: screens come straight from data tables, with tabs, lists, search, voice search and add/edit built in. It runs in a phone or computer browser.

What was hard

Layouts follow AppSheet's built-in views, so we could not recreate our sketched dashboard exactly. Some features in our design, such as QR scanning, were not built in the time available.

Prototype Tool: AppSheet

What helped

Quick to build: screens come straight from data tables, with tabs, lists, search, voice search and add/edit built in. It runs in a phone or computer browser.

What was hard

Layouts follow AppSheet's built-in views, so we could not recreate our sketched dashboard exactly. Some features in our design, such as QR scanning, were not built in the time available.

Limitations, Wizard of Oz & Hard-Coded Data

Limitations

A prototype to test the 3 tasks, not a finished system. Not built: QR scanning, role-based logins and separate labs, and automatic notifications.

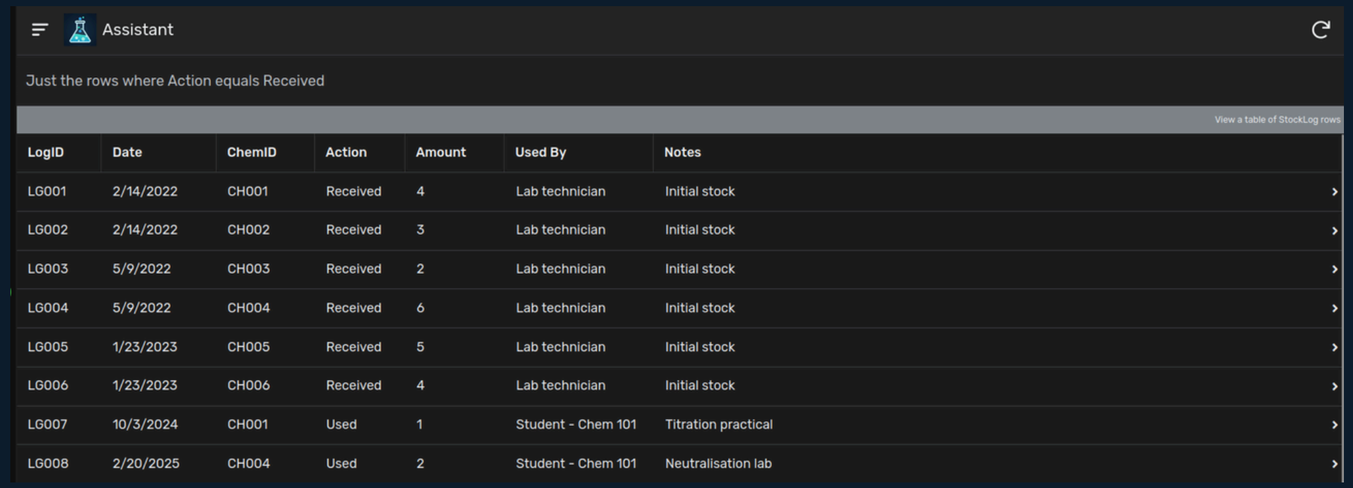
Wizard of Oz

The inventory data is simulated. We entered it by hand to stand in for a live inventory, and nothing is connected to real lab stock.

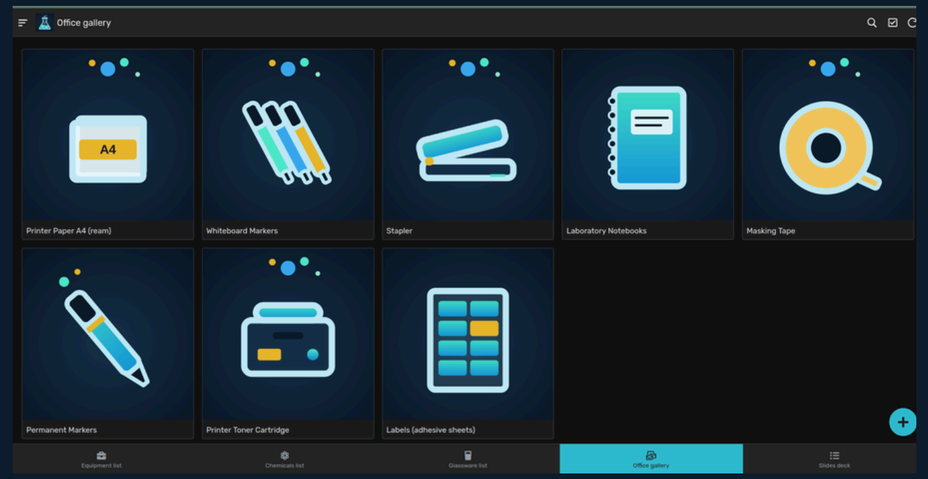
Hard-coded data

All equipment, chemicals, glassware, office supplies, slides and stock log entries are sample data entered in advance, so evaluators can try the tasks.

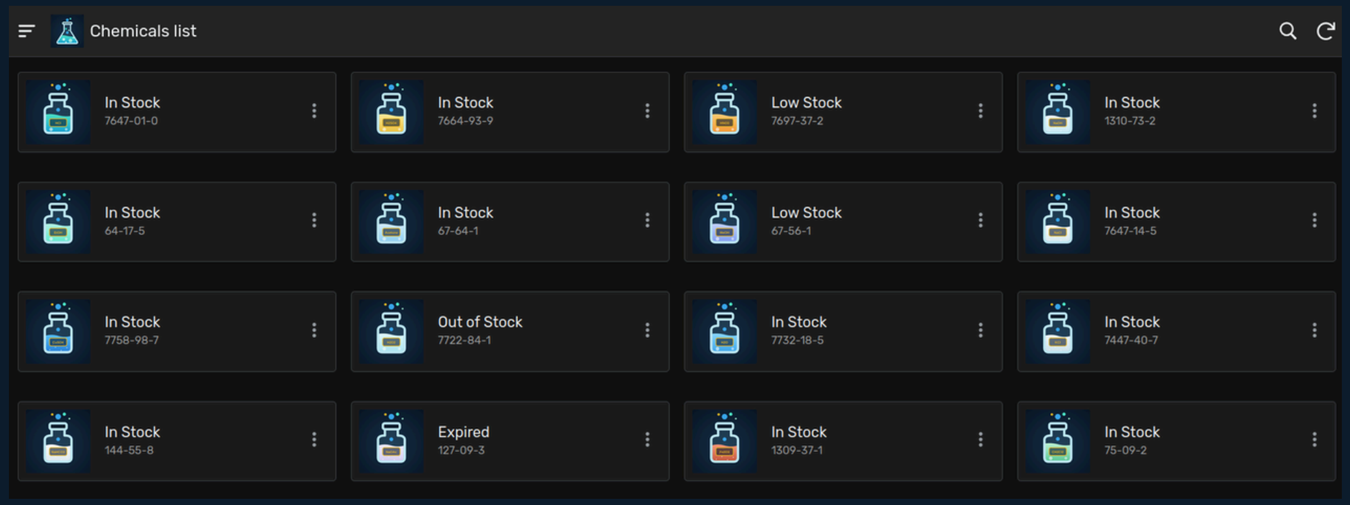
Additional Prototype Screenshots



Stock log: what was received or used, by whom and why



Office gallery tab



Chemicals list



Slides deck tab (microscope slides)